D02_API_Test_Environment

D02 — API test environment and functional captures

Branch: `cursor/d02-backend-cicd-38d2`
Purpose: Portal evidence item 2 — access to the API/test environment + functional captures
Safety: No production secrets, no real user credentials, no live JWTs in this document

1. How to access the test environment

Local disposable API test (verified on this branch)

Prerequisites: Node 24+, PostgreSQL listening locally.

```
        # Create disposable DB (example used for evidence)
        sudo -u postgres psql -c "CREATE USER grpd_test WITH PASSWORD
'grpd_test_local_only' SUPERUSER;"
        sudo -u postgres psql -c "CREATE DATABASE grpd_test OWNER
grpd_test;"

        export
DATABASE_URL='postgresql://grpd_test:grpd_test_local_only@127.0.0.1:5432/grpd_test
        export AUTH_JWT_SECRET='d02-local-test-jwt-secret-not-for-
prod'   # synthetic only

        npx prisma generate
        npx prisma migrate deploy

        # Automated journeys (auth success, 401, 403)
        npm run test:d02 -- --ci

        # Sanitized functional captures for the portal
        npx tsx scripts/d02-evidence-capture.ts
```

Artifacts written to `docs/funding/technical-evidence/D02/evidence/`:

- `functional-captures.json` — redacted request/response transcript
- `d02-integration.txt` — Jest D02 suite output
- `unit-tests.txt`, `typecheck-server.txt`, `migrations.txt`

CI test environment

GitHub Actions workflow: `.github/workflows/backend-verification.yml`

- Spins up Postgres 14 on port `55439`
- Runs `prisma migrate deploy`, unit tests, server typecheck, `npm run test:d02`
- Uploads `d02-ci-results/` as a workflow artifact

Hosted preview / production

- **Production was not redeployed** for D02.
- Existing production (`grpd30.vercel.app`) currently tracks an older email-auth preview branch and is out of scope for this evidence pack.
- To exercise a hosted **preview** of *this* branch later: open the PR → Vercel Preview URL → set Preview env `DATABASE_URL` (Neon test branch) + `AUTH_JWT_SECRET` (synthetic) in the Vercel project settings for Preview only. Do not paste those values into evidence files.

2. Functional captures (real results)

Source of truth: `evidence/functional-captures.json`
Captured at: see `capturedAt` inside that file.

Step	Method / path	Expected	Resu
Health	`GET /api/v1/health`	200, database ready	**PASS** (`health`
Register	`POST /api/v1/auth/email/register`	200 + tokens	**PASS** (`regist`
Unauthenticated	`GET /api/v1/me/memberships` (no Bearer)	401 `UNAUTHORIZED`	**PASS** (`unauth` `true`)
Create club	`POST /api/v1/clubs` (admin Bearer)	200	**PASS** (`create` `true`)
Invite	`POST /api/v1/clubs/:id/invite`	200 + code	**PASS** (`invite`
Join	`POST /api/v1/clubs/join-by-code`	200	**PASS** (`joinOk`
Role deny	`GET /api/v1/clubs/:id/roster` as member	403 `FORBIDDEN`	**PASS** (`memberRoste` `true`)
Role allow	`GET /api/v1/clubs/:id/roster` as admin	200 + members	**PASS** (`adminR` `true`)

Excerpt — unauthenticated rejection


```
{
  "step": "3_unauth_me",
  "method": "GET",
```

```
          "path": "/api/v1/me/memberships",
          "status": 401,
          "response": {
            "ok": false,
            "error": { "message": "Unauthorized", "code":
"UNAUTHORIZED" }
          }
        }
```


Excerpt — role access check

Member roster call returns `403 FORBIDDEN`; admin roster call returns `200` with members array (see captures `7_roster_member_forbidden` and `8_roster_admin_ok`).

3. Commands and results executed for this pack

```
npx prisma migrate deploy     → 6 migrations applied
(evidence/migrations.txt)
npm test -- --ci              → 7 suites / 69 tests passed
(evidence/unit-tests.txt)
npm run typecheck:server      → exit 0 (evidence/typecheck-server.txt)
npm run test:d02 -- --ci      → 2 suites / 11 tests passed
(evidence/d02-integration.txt)
npx tsx scripts/d02-evidence-capture.ts → all assertions true
(evidence/functional-captures.json)
```

4. Limitations

- Local disposable DB passwords shown above are **synthetic fixtures** for evaluators recreating the harness; they are not production credentials.
- Hosted Preview URL for this exact commit is created by Vercel when the PR is pushed; if Preview env vars are not yet configured, use the local/CI harness as the authoritative functional proof.
- PIN allowlist auth is available for fixture-only CI (`PIN_ALLOWLIST_JSON`); email auth is the self-serve path documented for testers.